

Sure, here are **five questions** in the same category (**Object-Oriented Programming (OOP) in Java using Inheritance and Polymorphism**):

1. University Management System:

A university wants to develop a system to manage different types of members using inheritance in Java. The base class `UniversityMember` should have attributes like `memberID`, `name`, and `department`. Two subclasses, `Student` and `Professor`, should inherit from `UniversityMember`.

- The `Student` class should have a `CGPA` attribute and a method `enrollCourse()`.
- The `Professor` class should have a `researchArea` attribute and a method `teachCourse()`.
- The system should create a list of `Student` and `Professor` objects, simulate course enrollments and teachings, and display member details.

2. Vehicle Rental System:

A rental service wants to develop a system to manage different types of vehicles. The base class `Vehicle` should have attributes like `vehicleID`, `brand`, and `rentalPricePerDay`. Two subclasses, `Car` and `Bike`, should inherit from `Vehicle`.

- The `Car` class should have an attribute `seats` and a method `rentCar()`.
- The `Bike` class should have an attribute `helmetIncluded` and a method `rentBike()`.
- The system should create a list of `Car` and `Bike` objects, simulate renting, and display rental details.

3. Online Shopping System:

An e-commerce platform wants to categorize products using OOP in Java. The base class `Product` should have attributes like `productID`, `name`, and `price`. Two subclasses, `Electronics` and `Clothing`, should inherit from `Product`.

- The `Electronics` class should have an attribute `warrantyPeriod` and a method `applyWarranty()`.
- The `Clothing` class should have an attribute `size` and a method `applyDiscount()`.
- The system should create a list of `Electronics` and `Clothing` objects, simulate purchases, and display order details.

4. Hospital Management System:

A hospital wants to track different types of staff and patients using inheritance in Java. The base class `HospitalMember` should have attributes like `ID`, `name`, and `age`. Two subclasses, `Doctor` and `Patient`, should inherit from `HospitalMember`.

- The `Doctor` class should have an attribute `specialization` and a method `diagnosePatient()`.
- The `Patient` class should have an attribute `disease` and a method `getTreatment()`.
- The system should create a list of `Doctor` and `Patient` objects, simulate patient diagnosis, and display medical records.

5. Bank Account Management System:

A bank wants to manage different types of accounts using OOP in Java. The base class `BankAccount` should have attributes like `accountNumber`, `holderName`, and `balance`. Two subclasses, `SavingsAccount` and `CurrentAccount`, should inherit from `BankAccount`.

- The `SavingsAccount` class should have an attribute `interestRate` and a method `applyInterest()`.
- The `CurrentAccount` class should have an attribute `overdraftLimit` and a method `withdraw()`.
- The system should create a list of `SavingsAccount` and `CurrentAccount` objects, simulate deposits and withdrawals, and display account details.